

Case Study

Energy Simulation data by eFilm software

PTT Head Office

Building	Window area (sq.ft.)	Total(sq.ft.)
Building 1	A=8611, B=8395,C=8395,D=7427	32,828
Building 2	A=36600, B=16468,C=11345,D=4205	68,618
Building 3	A=1550, B=2454,C=3358,D=1162	8,524



- Total window area = 109,970 sq.ft.
- Electricity bill in 2017 = 40,201,724 THB
- Energy cost = 3.76 THB/kWh.

Window film cost for simulation

Window film total cost = 13,196,400 THB.



Energy Simulation result by eFilm software

Building Energy Analysis Report

7/2/2018

Report generated by eFilm, an EnergyPlus based program.

Efilm version: v2.1.0.1

Simulation performed on: Saturday, June 30, 2018

EnergyPlus version: 8.0

Project Information

Title: PTT Building 1
Address: Vibhavadee Road
City/State/Zip: Bangkok,

Building Owner

Name:
Phone:
Address:
City/State/Zip:

Analyst

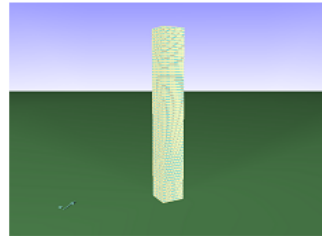
Name: Pichayapong Boonlue
Company: 3M
Address: Bangkok
City/State/Zip: Thailand

Architect

Name:
Phone:
Address:
City/State/Zip:

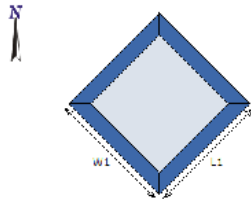
Building Information

Building Type: Office Buildings - Large
Building Shape: Rectangular
Orientation: North-East
Number of Floors: 59
Floor-to-Floor Height: 14.764 ft
Conditioned Area: 907,414.438 ft²
Vintage: 1978-1992
Category: Non-Residential



Dimensions

W1: 124.016 ft L1: 124.016 ft



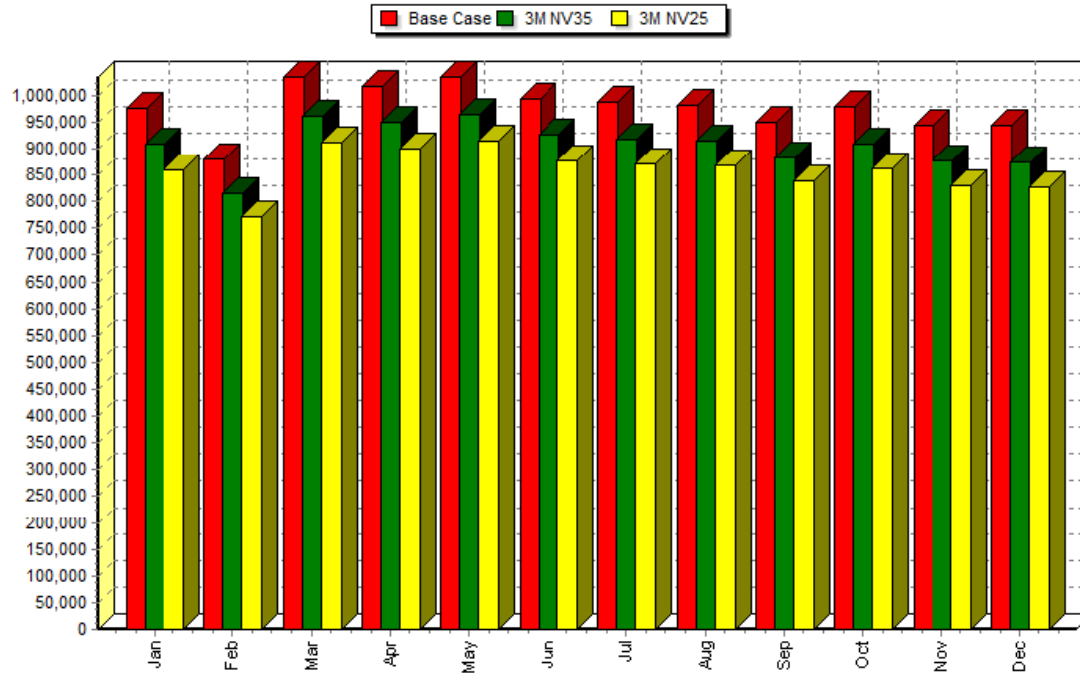
Results Summary

Option	Gas	Electricity	Total (Site)	Total (Source)	Savings
Energy Totals					
Base Case	4,614 MBTU	11,718,200 kWh	44,598 MBTU	131,509 MBTU	
3M NV35	3,753 MBTU	10,897,722 kWh	40,937 MBTU	121,713 MBTU	9,796 MBTU
3M NV25	3,201 MBTU	10,332,239 kWh	38,456 MBTU	115,007 MBTU	16,502 MBTU
Energy/Area					
Base Case	5.088 kBtu/ft²	12.914 kWh/ft²	49.182 kBtu/ft²	145.024 kBtu/ft²	
3M NV35	4.138 kBtu/ft²	12.01 kWh/ft²	45.144 kBtu/ft²	134.221 kBtu/ft²	10.803 kBtu/ft²
3M NV25	3.53 kBtu/ft²	11.386 kWh/ft²	42.408 kBtu/ft²	126.826 kBtu/ft²	18.198 kBtu/ft²
Peak Demand					
Base Case	5,017 kBtu/hr	2,354 kW			
3M NV35	4,760 kBtu/hr	2,231 kW			
3M NV25	4,568 kBtu/hr	2,146 kW			
Energy Cost					
Base Case	฿53,526	฿44,063,956	฿44,117,481		
3M NV35	฿43,530	฿40,978,712	฿41,022,243		฿3,095,239
3M NV25	฿37,127	฿38,852,326	฿38,889,454		฿5,228,027
Energy and HVAC Equipment Cost					
Base Case	฿62,421	฿44,272,584	฿44,335,004		
3M NV35	฿52,433	฿41,169,100	฿41,221,533		฿3,113,471
3M NV25	฿45,982	฿39,030,114	฿39,076,097		฿5,258,908

Energy Simulation result by eFilm software

Comparison Report

Monthly Electric Use [kWh]



Yearly Electric unit saving(kWh)

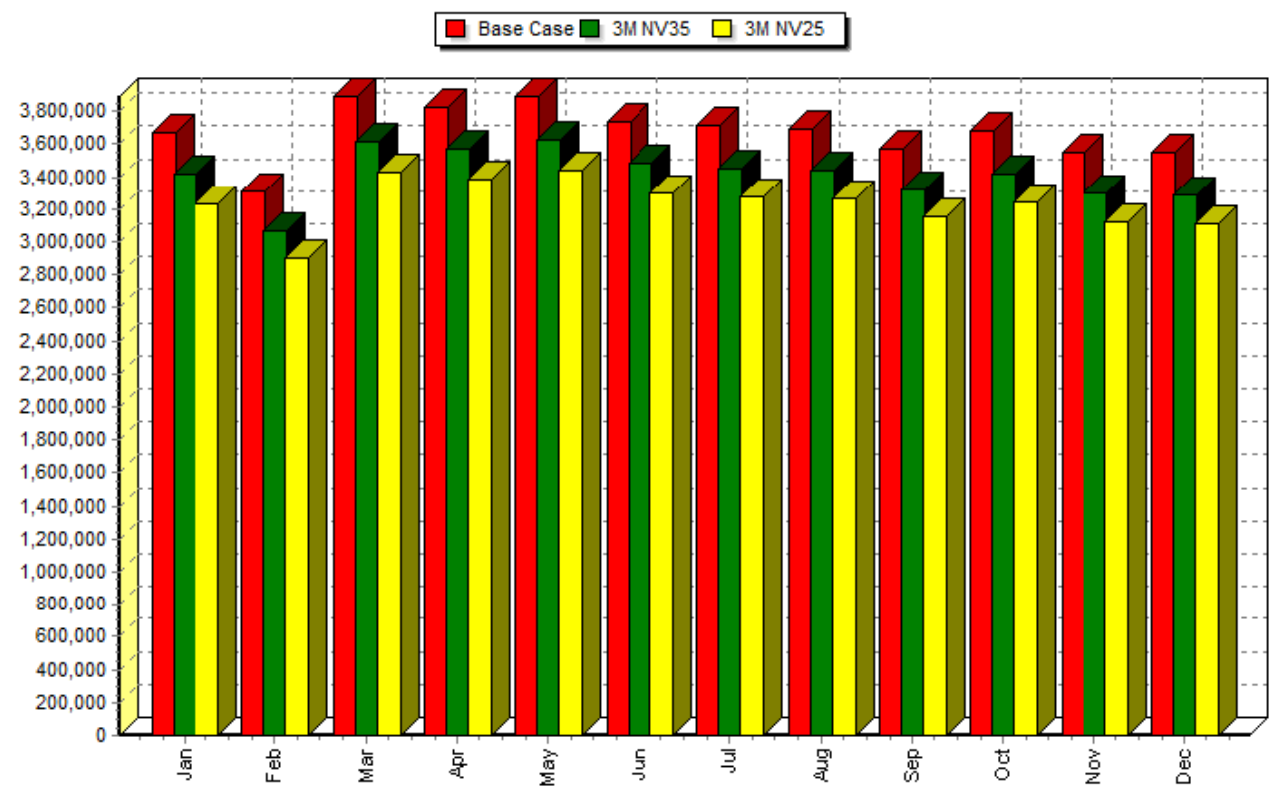
3M NV35 = 820,474 kWh.

3M NV25 = 1,385,962 kWh.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Base Case	977,170	879,171	1,034,412	1,017,838	1,033,511	993,262	987,290	981,622	950,016	977,517	943,839	942,550	11,718,198
3M NV35	907,849	815,855	961,959	948,383	963,362	924,991	918,105	913,791	883,614	909,203	876,615	873,999	10,897,724
3M NV35 - Savings	69,321	63,316	72,453	69,455	70,149	68,272	69,185	67,832	66,401	68,314	67,223	68,551	820,474
3M NV25	860,122	772,289	912,028	900,513	915,016	877,998	870,470	866,742	837,737	862,132	830,253	826,935	10,332,236
3M NV25 - Savings	117,048	106,882	122,384	117,325	118,494	115,264	116,819	114,881	112,279	115,386	113,586	115,615	1,385,962

Energy Simulation result by eFilm software

Monthly Electric Cost [฿]



Yearly Electric Cost saving(THB)

3M NV35 = 3,085,243 THB/year
3M NV25 = 5,211,629 THB/year

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Base Case	3,674,457	3,305,944	3,889,697	3,827,377	3,886,314	3,734,967	3,712,504	3,691,198	3,572,348	3,675,757	3,549,121	3,544,272	44,063,956
3M NV35	3,413,782	3,067,861	3,617,252	3,566,205	3,622,526	3,478,246	3,452,344	3,436,123	3,322,657	3,418,879	3,296,338	3,286,498	40,978,712
3M NV35 - Savings	260,675	238,082	272,445	261,172	263,788	256,721	260,160	255,074	249,691	256,878	252,784	257,774	3,085,243
3M NV25	3,234,318	2,904,040	3,429,501	3,386,201	3,440,740	3,301,542	3,273,233	3,259,206	3,150,146	3,241,873	3,122,005	3,109,520	38,852,326
3M NV25 - Savings	440,138	401,904	460,196	441,176	445,573	433,425	439,271	431,992	422,202	433,884	427,116	434,752	5,211,629

Energy Simulation result by Efilm software

Payback Period

Simple Payback

Option	Project Cost	Simple Payback Period
Base Case	฿0	
3M NV35	฿13,196,400	4.263 years
3M NV25	฿13,196,400	2.524 years

Payback Period for 3M NV35= 4.263 years.
Payback Period for 3M NV25= 2.524 years